校园实验室设备预约与借用管理系统

完整实体关系图谱

38个实体 / 328个字段 / 69条物理外键 / PostgreSQL

主键 · R01 → Parent.id · 外键与目标实体 · ? 可空 · UQ 单列唯一

实线：域内及主业务外键；共享责任主体引用以字段目标完整展开。

可编辑原文件可切换“共享引用”图层，检查全部69条连接。

A 身份与权限

共享身份、岗位和管理范围 · 6 实体

User A01 — 身份映射 — `core_user`

id	bigint	
username	varchar(150)	UQ
school_no	varchar(32)	UQ
password	varchar(128)	
last_login	timestamptz	?
is_superuser	boolean	
first_name	varchar(150)	
last_name	varchar(150)	
email	varchar(254)	
is_staff	boolean	
is_active	boolean	
date_joined	timestamptz	
display_name	varchar(60)	

13 fields · 主键 / 外键 / 完整字段

Role A02 — 业务角色 — `core_role`

id	bigint	
code	varchar(20)	UQ
name	varchar(40)	

3 fields · 主键 / 外键 / 完整字段

LabGrant A04 — 实验室管理范围 — `core_labgrant`

id	bigint	
user_id → User.id	bigint	A01
lab_id → Lab.id	bigint	B01
granted_at	timestamptz	

4 fields · 主键 / 外键 / 完整字段

UserRole A03 — 用户角色关联 — `core_userrole`

id	bigint	
user_id → User.id	bigint	A01
role_id → Role.id	bigint	A02
granted_by_id → User.id	bigint ?	A01
granted_at	timestamptz	

5 fields · 主键 / 外键 / 完整字段

QualificationType A05 — 培训资格类型 — `core_qualificationtype`

id	bigint	
code	varchar(32)	UQ
name	varchar(80)	

3 fields · 主键 / 外键 / 完整字段

UserQualification A06 — 用户培训资格 — `core_userqualification`

id	bigint	
user_id → User.id	bigint	A01
qualification_id → QualificationType.id	bigint	A05
granted_by_id → User.id	bigint ?	A01
issued_at	timestamptz	
expires_at	timestamptz ?	
revoked	boolean	

7 fields · 主键 / 外键 / 完整字段

B 资源与开放

资源主档、类型明细和日历 · 7 实体

Lab B01 — 实验室 — `core_lab`

id	bigint	
code	varchar(32)	UQ
name	varchar(80)	
location	varchar(120)	
description	text	
is_active	boolean	

6 fields · 主键 / 外键 / 完整字段

OpeningPeriod B06 — 日间开放时段 — `core_openingperiod`

id	bigint	
lab_id → Lab.id	bigint	B01
resource_id → Resource.id	bigint ?	B02
weekday	smallint	
starts_at	time	
ends_at	time	
closed_all_day	boolean	
valid_from	date	
valid_to	date ?	

9 fields · 主键 / 外键 / 完整字段

Accessory B05 — 设备配件档案 — `core_accessory`

id	bigint	
resource_id → Resource.id	bigint	B02
code	varchar(32)	
name	varchar(80)	
quantity	integer	
is_active	boolean	

6 fields · 主键 / 外键 / 完整字段

CalendarException B07 — 单日开放例外和本 — `core_calendarexception`

id	bigint	
lab_id → Lab.id	bigint	B01
actor_id → User.id	bigint	A01
day	date	
closed	boolean	
starts_at	time ?	
ends_at	time ?	
active	boolean	
reason	text	
created_at	timestamptz	

10 fields · 主键 / 外键 / 完整字段

Resource B02 — 可预约资源 — `core_resource`

id	bigint	
lab_id → Lab.id	bigint	B01
code	varchar(40)	UQ
name	varchar(100)	
kind	varchar(16)	
description	text	
service_status	varchar(20)	
manual_hold	varchar(32)	
created_at	timestamptz	
updated_at	timestamptz	

16 fields · 主键 / 外键 / 完整字段

EquipmentDetail B03 — 设备属性 — `core_equipmentdetail`

id	bigint	
resource_id → Resource.id	bigint	UQ B02
qualification_id → QualificationType.id	bigint ?	A05
access_room_id → Resource.id	bigint ?	B02
category	varchar(40)	
risk	varchar(20)	
physical_status	varchar(16)	
portable	boolean	
handover_location	varchar(120)	

9 fields · 主键 / 外键 / 完整字段

RoomDetail B04 — 场地属性 — `core_roomdetail`

id	bigint	
resource_id → Resource.id	bigint	UQ B02
location_label	varchar(160)	
capacity	integer	
space_category	varchar(40)	
facilities	text	
safety_note	text	

7 fields · 主键 / 外键 / 完整字段

C 预约与规则

申请、最终决定和规则副本 · 6 实体

Reservation C01 — 资源预约 — `core_reservation`

id	bigint	
applicant_id → User.id	bigint	A01
resource_id → Resource.id	bigint	B02
code	varchar(40)	UQ
starts_at	timestamptz	
ends_at	timestamptz	
purpose	text	
participants	integer	
status	varchar(16)	
approval_mode	varchar(16)	
request_key	uuid	
request_fingerprint	varchar(64)	
policy_snapshot	jsonb	
reason	text	
version	integer	
review_required	boolean	
review_reason	text	
arrived_at	timestamptz ?	
late_start_until	timestamptz ?	
created_at	timestamptz	

20 fields · 主键 / 外键 / 完整字段

ApprovalDecision C02 — 自动或人工确认决定 — `core_approvaldecision`

id	bigint	
reservation_id → Reservation.id	bigint	UQ C01
actor_id → User.id	bigint ?	A01
mode	varchar(16)	
approved	boolean	
reason	text	
decided_at	timestamptz	

7 fields · 主键 / 外键 / 完整字段

ReservationReview C06 — 预约复核历史 — `core_reservationreview`

id	bigint	
reservation_id → Reservation.id	bigint	C01
actor_id → User.id	bigint	A01
accepted	boolean	
reason	text	
snapshot	jsonb	
created_at	timestamptz	

7 fields · 主键 / 外键 / 完整字段

Policy C04 — 当前业务参数 — `core_policy`

id	bigint	
version	integer	
window_days	integer	
slot_minutes	integer	
max_hours	integer	
equipment_quota	integer	
room_quota	integer	
simultaneous_equipment	integer	
grace_minutes	integer	
mock_delivery_failure	boolean	
updated_at	timestamptz	

11 fields · 主键 / 外键 / 完整字段

PolicyRevision C05 — 参数历史版本 — `core_policyrevision`

id	bigint	
policy_id → Policy.id	bigint	C04
actor_id → User.id	bigint ?	A01
version	integer	
data	jsonb	
reason	text	
created_at	timestamptz	

7 fields · 主键 / 外键 / 完整字段

ReviewNote C03 — 审核补充记录 — `core_reviewnote`

id	bigint	
reservation_id → Reservation.id	bigint	C01
actor_id → User.id	bigint	A01
kind	varchar(9)	
content	text	
evidence_ref	varchar(240)	
created_at	timestamptz	

7 fields · 主键 / 外键 / 完整字段

D 现场使用

设备交接与场地实际使用 · 5 实体

Loan D01 — 设备借出事实 — `core_loan`

id	bigint	
reservation_id → Reservation.id	bigint	UQ C01
operator_id → User.id	bigint	A01
code	varchar(40)	UQ
arrived_at	timestamptz	
service_started_at	timestamptz	
handed_over_at	timestamptz	
due_at	timestamptz	
tag_snapshot	varchar(40)	
condition_out	text	

10 fields · 主键 / 外键 / 完整字段

ReturnRecord D02 — 设备归还事实 — `core_returnrecord`

id	bigint	
loan_id → Loan.id	bigint	UQ D01
operator_id → User.id	bigint	A01
code	varchar(40)	UQ
returned_at	timestamptz	
damaged	boolean	
missing	boolean	
fault	boolean	
note	text	

9 fields · 主键 / 外键 / 完整字段

RoomUse D04 — 场地实际使用与检查 — `core_roomuse`

id	bigint	
reservation_id → Reservation.id	bigint	UQ C01
started_by_id → User.id	bigint	A01
ended_by_id → User.id	bigint ?	A01
arrived_at	timestamptz	
started_at	timestamptz	
actual_participants	integer	
start_note	text	
ended_at	timestamptz ?	
end_note	text	
abnormal	boolean	

11 fields · 主键 / 外键 / 完整字段

HandoverItem D03 — 配件交接快照 — `core_handoveritem`

id	bigint	
loan_id → Loan.id	bigint	D01
accessory_id → Accessory.id	bigint ?	B05
code_snapshot	varchar(32)	
name_snapshot	varchar(80)	
expected_qty	integer	
out_qty	integer	
out_condition	varchar(160)	
return_qty	integer ?	
return_condition	varchar(160)	

10 fields · 主键 / 外键 / 完整字段

Coordination D06 — 预约协调凭据 — `core_coordination`

id	bigint	
reservation_id → Reservation.id	bigint	C01
actor_id → User.id	bigint	A01
action	varchar(20)	
reason	text	
old_state	jsonb	
new_state	jsonb	
created_at	timestamptz	

8 fields · 主键 / 外键 / 完整字段

E 异常与维修

证据、工单和独立验收 · 5 实体

Incident E01 — 资源异常事实 — `core_incident`

id	bigint	
resource_id → Resource.id	bigint	B02
reporter_id → User.id	bigint	A01
reservation_id → Reservation.id	bigint ?	C01
confirmed_by_id → User.id	bigint ?	A01
code	varchar(40)	UQ
kind	varchar(20)	
description	text	
blocks_service	boolean	
occurred_at	timestamptz	
reported_at	timestamptz	
status	varchar(16)	
conclusion	text	
resolved_at	timestamptz ?	

14 fields · 主键 / 外键 / 完整字段

Evidence E02 — 异常证据索引 — `core_evidence`

id	bigint	
incident_id → Incident.id	bigint	E01
uploaded_by_id → User.id	bigint	A01
file	varchar(100)	
original_name	varchar(160)	
sha256	varchar(64)	
uploaded_at	timestamptz	

7 fields · 主键 / 外键 / 完整字段

IncidentResponse E05 — 异常确认与评议记录 — `core_incidentresponse`

id	bigint	
incident_id → Incident.id	bigint	E01
actor_id → User.id	bigint	A01
kind	varchar(20)	
content	text	
created_at	timestamptz	

6 fields · 主键 / 外键 / 完整字段

WorkOrder E03 — 维修工单 — `core_workorder`

id	bigint	
incident_id → Incident.id	bigint	E01
created_by_id → User.id	bigint	A01
assigned_to_id → User.id	bigint ?	A01
accepted_by_id → User.id	bigint ?	A01
code	varchar(40)	UQ
status	varchar(16)	
priority	varchar(12)	
result	text	
finished_at	timestamptz ?	
accepted_at	timestamptz ?	
created_at	timestamptz	

12 fields · 主键 / 外键 / 完整字段

WorkOrderEvent E04 — 维修及验收过程 — `core_workorderevent`

id	bigint	
work_order_id → WorkOrder.id	bigint	E03
actor_id → User.id	bigint	A01
old_status	varchar(24)	
new_status	varchar(24)	
note	text	
created_at	timestamptz	

7 fields · 主键 / 外键 / 完整字段

F 审计与通知

状态区间和可追踪投递 · 7 实体

AuditEvent F01 — 不可重置审计事件 — `core_auditevent`

id	bigint	
actor_id → User.id	bigint ?	A01
lab_id → Lab.id	bigint ?	B01
dedupe_key	varchar(160)	? UQ
actor_kind	varchar(8)	
entity_type	varchar(40)	
entity_id	bigint	
action	varchar(40)	
old_state	jsonb	
new_state	jsonb	
reason	text	
request_id	uuid	
created_at	timestamptz	

13 fields · 主键 / 外键 / 完整字段

ServicePeriod F02 — 资源服务状态区间 — `core_serviceperiod`

id	bigint	
resource_id → Resource.id	bigint	B02
audit_id → AuditEvent.id	bigint	UQ F01
status	varchar(20)	
starts_at	timestamptz	
ends_at	timestamptz ?	

6 fields · 主键 / 外键 / 完整字段

MockReceipt F06 — 模拟投递主要凭据 — `core_mockreceipt`

id	bigint	
delivery_id → Delivery.id	bigint	UQ F04
received_at	timestamptz	

3 fields · 主键 / 外键 / 完整字段

Notification F03 — 接收人通知 — `core_notification`

id	bigint	
audit_id → AuditEvent.id	bigint	F01
recipient_id → User.id	bigint	A01
code	varchar(40)	UQ
kind	varchar(32)	
content	text	
created_at	timestamptz	
read_at	timestamptz ?	

9 fields · 主键 / 外键 / 完整字段

Delivery F04 — 退回投递任务 — `core_delivery`

id	bigint	
notification_id → Notification.id	bigint	F03
channel	varchar(16)	
status	varchar(16)	
retry_count	smallint	
next_attempt_at	timestamptz	
locked_until	timestamptz ?	
last_error	text	

8 fields · 主键 / 外键 / 完整字段

DeliveryAttempt F05 — 投递尝试 — `core_deliveryattempt`

id	bigint	
delivery_id → Delivery.id	bigint	F04
attempt_no	smallint	
outcome	varchar(12)	
detail	text	
attempted_at	timestamptz	

6 fields · 主键 / 外键 / 完整字段

JobHeartbeat F07 — 后台任务健康记录 — `core_jobheartbeat`

id	bigint	
name	varchar(40)	UQ
last_success	timestamptz ?	
last_error_at	timestamptz ?	
error_code	varchar(80)	

5 fields · 主键 / 外键 / 完整字段

G 资源安全信息

适用于管理单元或具体资源。具来源与复核版本的安全信息 · 2 实体

SafetyProcedure G01 — 安全信息版本 — `core_safetyprocedure`

id	bigint	
lab_scope_id → Lab.id	bigint ?	B01
resource_id → Resource.id	bigint ?	B02
created_by_id → User.id	bigint	A01
reviewed_by_id → User.id	bigint ?	A01
published_by_id → User.id	bigint ?	A01
version	integer	
title	varchar(160)	
risk_categories	text	
applicable_users	text	
prerequisites	text	
required_ppe	text	
pre_use_checks	text	
operating_cautions	text	
prohibited_actions	text	
incident_route	text	
source_title	varchar(200)	
source_reference	varchar(500)	
source_version	varchar(80)	
source_effective_date	date ?	
effective_from	date ?	
review_due_date	date ?	
requires_acknowledgement	boolean	
status	varchar(20)	
submitted_at	timestamptz ?	
reviewed_at	timestamptz ?	
review_note	text	
published_at	timestamptz ?	
change_reason	text	
created_at	timestamptz	

30 fields · 主键 / 外键 / 完整字段

SafetyAcknowledgement G02 — 安全信息阅读确认 — `core_safetyacknowledgement`

id	bigint	
procedure_id → SafetyProcedure.id	bigint	G01
user_id → User.id	bigint	A01
acknowledged_at	timestamptz	

4 fields · 主键 / 外键 / 完整字段

阅读顺序：资源 → 预约 → 现场使用 → 异常维修；身份、规则与审计为共用支撑。

实体代码用于跨域定位；R01–R69对应完整关系索引。实体、字段及外键由实际模型导出。

关键约束

排带签不占资源；确认与使用中的重叠区间互斥。

设备与场地分别记录实际使用；审计逻辑引用不伪装为外键。

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